Materials Engineer (Starlink)

SpaceX · Bastrop, TX

Job ID (this archive)	MFG-065
Employer	SpaceX
Location	Bastrop, TX
Career stage	New grad / early career
Location group	Texas (outside DFW)
Major (archive tag)	Materials Science & Engineering
Sector area	Electronics & semiconductor manufacturing
Industry (LinkedIn)	Aviation and Aerospace Component Manufacturing
Pay listed	Not listed
Stated rules	U.S. citizen / ITAR
Posted / closes	2026-10-29 · 2027-01-02
Status when checked	Accepting applications on LinkedIn (Nov 1, 2026)
Source page	https://www.linkedin.com/jobs/view/4449444149/
Captured	2026-11-01

Full posting text as it appeared on LinkedIn on Nov 1, 2026, saved for ECS 1100. Postings change and close often; open the source page above for the current version before applying.

At a glance

What you would do

Engineer selecting, qualifying and supporting the consumable materials used to make solar cells for satellites, working with suppliers and production teams.

Who can apply

- **Majors named:** Materials Science, Chemical Engineering, Any STEM degree
- **Degree and experience:** Bachelor's in materials science, chemical engineering or a STEM field; 1+ years materials engineering experience

Skills mentioned

Teamwork & collaboration · Communication · Problem solving & root cause · Safety · Supply chain, planning & logistics

Full posting text

SpaceX was founded under the belief that a future where humanity is out exploring the stars is fundamentally more exciting than one where we are not. Today SpaceX is actively developing the technologies to make this possible, with the ultimate goal of enabling human life on Mars.

MATERIALS ENGINEER (STARLINK)

Starlink is building the world's largest satellite constellation to deliver fast, reliable internet to people who need it most—and powering that constellation starts on the factory floor. We are insourcing solar cell manufacturing end-to-end, from crystal growth and ingot production through wafering and cell fabrication to drive performance, cost, and scale at every step. High-efficiency, high-reliability cells made in-house are the backbone of the Starlink network: more photons in means more data out to customers around the world.

As a Materials Engineer, you will own the critical consumables that keep our ingot, wafer, and solar cell manufacturing lines running. You will take end-to-end ownership of materials used in production—partnering with supply chain on sourcing, driving material qualification, and handing off proven materials into high-volume manufacturing. We are looking for a general Materials Engineer who is excited about metals, polymers, crystals, and process consumables; solar experience is a plus, not a requirement. If you are passionate about solving hard problems, anchor your engineering judgment with first principles thinking, and are relentless about acting with urgency, this role is for you.

Responsibilities

- Own consumable materials used across ingot, wafer, and/or solar cell manufacturing lines—from initial sourcing and technical evaluation through qualification, production hand-off, and ongoing support.
- Partner closely with supply chain to identify, evaluate, and qualify suppliers for critical process materials.
- Define material requirements, specifications, and acceptance criteria in collaboration with process, quality, and production engineering teams.
- Plan and execute material qualification campaigns, including characterization, process trials, and production-readiness assessments before hand-off to manufacturing.
- Drive root cause to first principles and implement durable corrective actions.
- Build deep technical understanding of material chemistry, microstructure, and process interactions for metals, polymers, crystals, and other consumables used in high-volume manufacturing.
- Identify opportunities to improve material performance, cost, yield, and reliability through alternative materials, dual-sourcing strategies, or process changes.
- Document qualification results, material specifications, and best practices.
- Communicate status, risks, and recommendations clearly to cross-functional partners and leadership.
- Champion a culture of safety and quality in all materials handling, testing, and production support activities.

Basic Qualifications

Bachelor's degree in materials science, chemical engineering, or a STEM field.1+ years of experience in materials engineering, materials selection, process materials, or manufacturing materials support.

Preferred Skills And Experience

- Master’s degree or PhD in materials science, chemical engineering, or a related field.
- Hands-on experience with metals, polymers, crystals, ceramics, coatings, or other process consumables in a manufacturing or R&D environment.
- Experience with material qualification, supplier development, or transitioning materials from development into production.
- Familiarity with materials characterization techniques (e.g., SEM-EDS, XRD, FTIR, optical microscopy, mechanical or chemical property testing).
- Experience working in a high-volume manufacturing environment and partnering with production, quality, and supply chain teams.
- Strong first-principles problem solving and a demonstrated ability to dig deep on ambiguous materials challenges.
- Passion for building, learning quickly, and owning outcomes end-to-end; thrives in a fast-paced, scrappy environment with tight timelines.
- Effective written and interpersonal communication skills; able to convey technical concepts across all levels of the organization.

Additional Requirements

- Willingness and ability to travel up to 25% time.
- Ability to lift up to 25 lbs unassisted.
- Ability to stand/walk for extended periods on the manufacturing floor.
- Willingness to work extended hours and weekends as needed.

Itar Requirements

To conform to U.S. Government export regulations, applicant must be a (i) U.S. citizen or national, (ii) U.S. lawful, permanent resident (aka green card holder), (iii) Refugee under 8 U.S.C.1157, or (iv) Asylee under 8 U.S.C.1158, or be eligible to obtain the required authorizations from the U.S. Department of State. Learn more about the ITAR here.

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Applicants wishing to view a copy of SpaceX's Affirmative Action Plan for veterans and individuals with disabilities, or applicants requiring reasonable accommodation to the application/interview process should reach out to EEOCompliance@spacex.com.